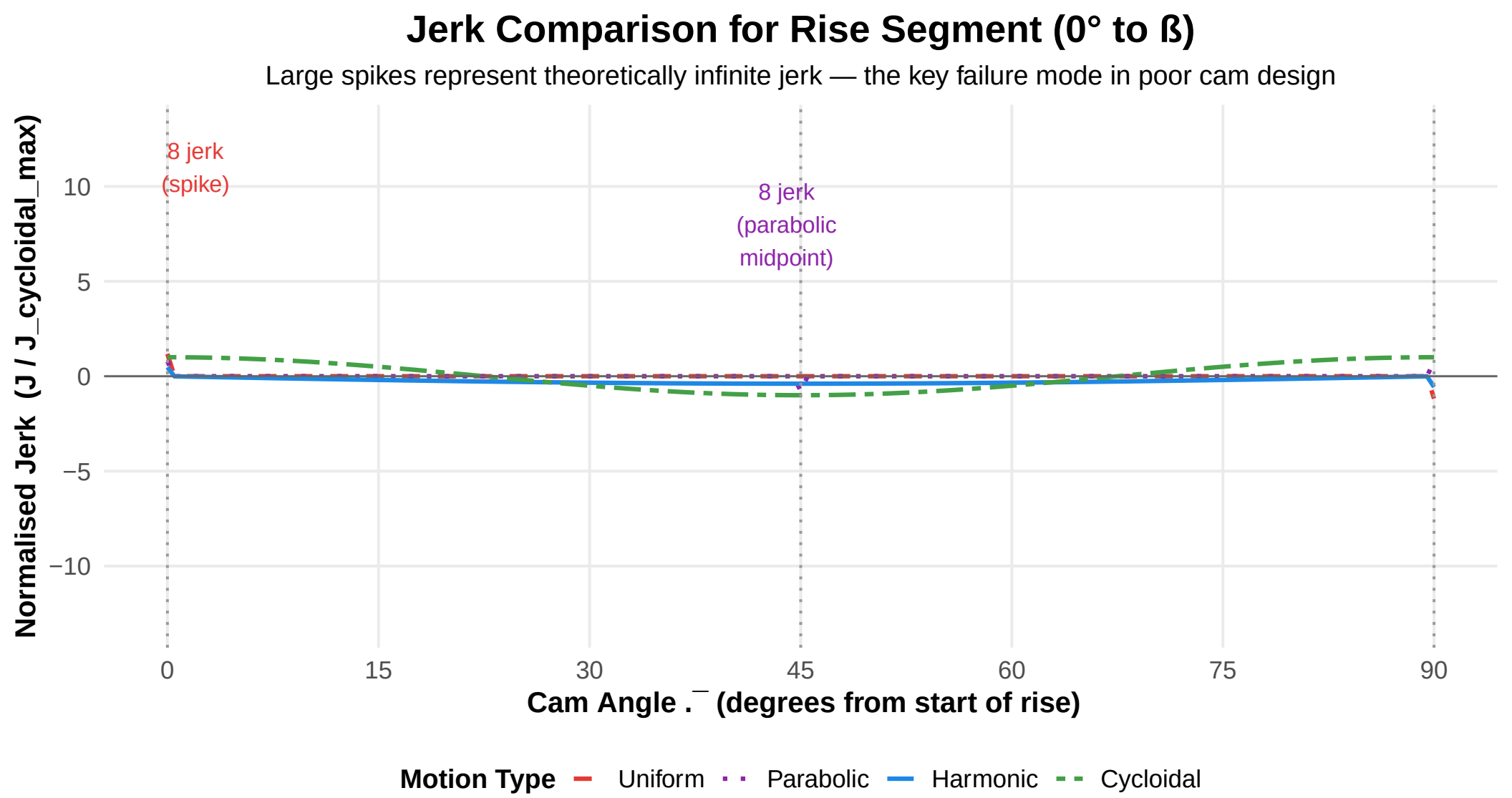
Jerk Comparison for Rise Segment (0° to ß)
Large spikes represent theoretically infinite jerk — the key failure mode in poor cam design
8 jerk
(spike)
8 jerk
(parabolic
midpoint)
Normalised Jerk (J / J_cycloidal_max)
10
5
0
−5
−10
0
15
30
45
60
75
90
Cam Angle .¯ (degrees from start of rise)
Motion Type
Uniform
Parabolic
Harmonic
Cycloidal